

CS 5330 HW 7

1. What do the extrinsic parameters of a camera specify?

The parameters of a camera that are unique to each image, and which change whenever you change the position and rotation of the camera. In fact, the camera position and orientation ARE extrinsic parameters of a camera

2. What do the intrinsic parameters of a camera specify?

These parameters do not change between images taken by the camera. These parameters are often fixed (unless a camera allows changing of the focal length for example). These include focal length, pixel size, and the center of the image / focal point.

3. What is the difference between Camera Euclidean coordinates and Image Euclidean coordinates?

Image Euclidean coordinates are 2D with the origin centered on the middle of the image, Camera Euclidean coordinates are the 3D with the origin centered on the camera. Image Euclidean is Camera Euclidean with all world points projected onto the image plane. Units are still the same for both.

4. What is the difference between Image Euclidean coordinates and Image Affine coordinates?

Image Affine coordinates (aka pixel coordinates) is the Image Euclidean coordinates, but with the axes flipped and translated (based on the height and width of the camera) such that all coords are positive, the new origin is in the top left of the image plane, and units are pixels.

5. Why do AR applications ask you to move your phone side to side in a circle during the setup process?

It needs to calibrate the intrinsics / extrinsics by having the phone do rotations and translations to capture images from different viewpoints. This lets it identify similar points in each image, assign world coords to points on the target, and do nonlinear optimization to get intrinsic / extrinsic parameters.

6. Given a focal length of 10mm, a baseline of 100mm, and pixel size of 0.01mm.

- a. What is the distance at which the disparity between two images is less than half a pixel?

The disparity equation is

$$d = \frac{fb}{z}$$

Solving for z (distance)

$$z = \frac{fb}{d} = \frac{10 * 100}{0.005} = 200000mm = 200m$$

- b. What is the distance at which the disparity between two images is two pixels?

$$z = \frac{fb}{d} = \frac{10 * 100}{0.02} = 50000mm = 50m$$

7. After rectifying the images for a stereo pair, if a feature in the left image is at pixel (x, y), where do you have to search in the right image for the corresponding feature? Is there an upper limit on how many locations you need to search?

You have to search to the left of those coordinates in the right image for the same feature. There is not really an upper limit on how many location you need to search, as finding the corresponding feature in both images can be very tricky and may require looking through all pixels to the left of those coordinates and doing some sum squared distance or other distance metric. There can be complications with all pixels in that part of the image being the same color, or having a repeating pattern, which multi-resolution matching may fix, but either way these approaches don't have a guaranteed "upper limit" where it is guaranteed to find the right match.

8. The horopter is the set of points in space that fall on the same pixel in a stereo pair (disparity zero). Look it up on wikipedia.
- a. What is the horopter for a stereo pair with parallel optical axes?

The horopter is a point located infinitely far from the camera. The axes never intersect

- b. Cat's eyes are not parallel, but are oriented so the horopter is a set of points in the world between their paws. Why do you think evolution gave them that trait?

This would let the cat have better depth perception when chasing prey like mice for example.

9. You are using the 8-point algorithm to estimate the Fundamental matrix. However, your data has about 10% incorrect matches. If you use the RANSAC algorithm to find the best F matrix, how many times would you need to run the inner loop in order to have 99% confidence in finding a good model? What if the number of outliers is 20%? Hint: read my lecture notes before answering this problem.

For 10% outliers (10% of matching pairs between features in image A and B are off by some amount of pixels), and desired 99% confidence, we can compute T as

$p = 0.99$ (99% confidence), $E = 0.10$ (10% outliers), $s = 8$ (8-point model)

$$T = \frac{\log(1 - p)}{\log(1 - (1 - E)^s)}$$

$$T = \frac{\log(1 - 0.99)}{\log(1 - (1 - 0.10)^8)} = 8.18$$

And for 20% outliers:

$$T = \frac{\log(1 - 0.99)}{\log(1 - (1 - 0.20)^8)} = 25.07$$